

Jewelry Production

Create a production plan for your aunt's necklace production, to maximize her income.

Problem

- Maximize production profit for your aunt.

Sets

- $n \in Necklaces = \{1, 2, 3, 4, 5\}$
- $m \in Machine = \{1, 2, 3\}$

Parameters

- $Profit_n$: Profit for selling necklace n
- $MachineTime_{m,n}$: Number of minutes needed for each necklace n on machine m
- $AssemblyTime_n$: Number of minutes needed to assemble necklace n
- $DayMinutes$: No of minutes pr. day
- $NoAssemblyWorkers$: No of assembly workers employed

Decision variables

- Number of necklaces of type n produced: $x_n \geq 0$.

Model

Objective:

- Maximize the necklace production profit:

$$\sum_n Profit_n \cdot x_n$$

Constraints:

- Limit the production to the machine capacity:

$$\sum_n MachineTime_{m,n} \cdot x_n \leq DayMinutes \quad \forall m$$

- Limit the production according to the assembly capacity:

$$\sum_n AssemblyTime_n \cdot x_n \leq DayMinutes \cdot NoAssemblyWorkers$$

The full model in Julia/JuMP, available with the name

`JewelleryProduction.jl`

from the book web-site, is given below:

```

*****
# Jewellery Production assignment, LP
using JuMP
using HiGHS
*****

*****
# Data
Necklaces=["1","2","3","4","5"]
N=length(Necklaces)
Machines=["1","2","3"]
M=length(Machines)
Profit=[50 45 85 60 55]
AssemblyTime=[12 3 11 9 6]
MachineTime=[
    7 0 0 9 0;
    5 7 11 0 5;
    0 3 8 15 3]
DayMinutes=60*7.5
NoAssemblyWorkers=2
*****

*****

```

```

# Model
JP = Model(HiGHS.Optimizer)

@variable(JP,x[n=1:N]>=0)

# maximize profit
@objective(JP, Max, sum( Profit[n]*x[n] for n=1:N) )

# Machine capacity
@constraint(JP, [m=1:M],
    sum( MachineTime[m,n]*x[n] for n=1:N) <= DayMinutes
)

# Assembly capacity
@constraint(JP,
    sum(AssemblyTime[n]*x[n] for n=1:N) <= DayMinutes*NoAssemblyWorkers
)

#####

#####
# Solve
solution = optimize!(JP)
println("Termination status: $(termination_status(JP))")
#####

#####
if termination_status(JP) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(JP))")
    for n=1:N
        println("Production of necklace $(Necklaces[n]) : ", value(x[n]))
    end
else
    println("No optimal solution available")
end
#####

```

In a market economy, you cannot force people to buy your products. Hence, it is necessary to consider the demand as a constraint on how many necklaces can be sold. This is a simple extension, but as we are making the problem more constrained, we will expect the objective value to be the same or worse (lower).

More Parameters

- $Demand_n$: The maximal number of necklace n which can be sold.

Additional Constraints:

- Added constraint to limit sales according to the demand:

$$x_n \leq Demand_n \quad \forall n$$

The full model in Julia/JuMP, available with the name

`JewelleryProduction2.jl`

from the book web-site, is given below:

```
*****
# Jewellery Production 2 assignment, LP
using JuMP
using HiGHS
*****

# Data
Necklaces=["Necklace 1","Necklace 2","Necklace 3","Necklace 4","Necklace 5"]
N=length(Necklaces)
Machines=["Machine 1","Machine 2","Machine 3"]
M=length(Machines)
Profit=[50 45 85 60 55]
AssemblyTime=[12 3 11 9 6]
MachineTime=[
    7 0 0 9 0;
    5 7 11 0 5;
    0 3 8 15 3]
DayMinutes=60*7.5
NoAssemblyWorkers=3
Demand=[25 10 12 15 60]
*****

# Model
JP = Model(HiGHS.Optimizer)

@variable(JP,x[n=1:N]>=0)

# maximize profit
@objective(JP, Max, sum( Profit[n]*x[n] for n=1:N) )

# Machine capacity
```

```

@constraint(JP, [m=1:M],
            sum( MachineTime[m,n]*x[n] for n=1:N) <= DayMinutes
            )

# Assembly capacity
@constraint(JP,
            sum(AssemblyTime[n]*x[n] for n=1:N) <= DayMinutes*NoAssemblyWorkers
            )

# Demand limits
@constraint(JP, [n=1:N],
            x[n] <= Demand[n]
            )

#####

#####
# Solve
solution = optimize!(JP)
println("Termination status: $(termination_status(JP))")
#####

#####
if termination_status(JP) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(JP))")
    for n=1:N
        println("Production of necklace $(Necklaces[n]) : ", value(x[n]))
    end
else
    println("No optimal solution available")
end
#####

#####
# Jewellery Production 2 assignment, LP, Modified print
if termination_status(JP) == MOI.OPTIMAL

```

To evaluate whether your aunt should hire another assembly worker there are two options: 1) Increase the value of *NOAssemblyWorkers* to three in your model and resolve it. Inspect whether this improves the solution. Alternatively, 2) Evaluate the left hand side of the assembly time constraint to see if it is satisfied with equality or whether there is unused capacity. In the latter case there is no need to hire additional workers. This can be done by modifying the print as follows to reveal that only 820 minutes out of the available 1350 are used.

```

#####
# Jewellery Production 2 assignment, LP, Modified print
if termination_status(JP) == MOI.OPTIMAL

```

```

println("Optimal objective value: $(objective_value(JP))")
for n=1:N
    println("Production of necklace $(Necklaces[n]) : ", value(x[n]))
end
# Access the need for additional workers
println("Assembly time used: $(sum(AssemblyTime[n]*value(x[n]) for n=1:N))
        out of $(DayMinutes*NoAssemblyWorkers)")
else
    println("No optimal solution available")
end
#####

```